

ANDREW ROOT

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PROFESSIONAL EXPERIENCE

Founding Engineer — *Daentra (Startup)* — San Francisco, CA 08/25 - Present

- Developed and deployed an end-to-end vision/audio system including sensors, edge devices, and Cloud Infrastructure. Deployed in multiple dental operatory rooms, across 2 offices.
- Researching and prototyping depth and pose estimation methods for key object tracking in video: evaluating modern models and implementing custom solutions.
- Developed a novel solution for synchronization of a machine vision camera and a flickering light that has a varying frequency.

Research Lab Assistant — *Oakland University* — Oakland County, MI 05/24 - 05/25

- Developed a real-time machine learning pipeline for foreign-object detection in video data for an automated security vehicle system (with Dataspeed Inc.).

Undergraduate Researcher — *University of Wisconsin Eau Claire* — Eau Claire, WI 09/23 - 12/24

- Conducted cyberbullying detection research by investigating sources of bias in ML datasets, using cross-dataset machine learning evaluation on the university HPC cluster, and mentored younger students.

Software Engineer Intern — *Midwest Manufacturing (Menards Inc.)* — Eau Claire, WI 05/23 - 05/24

- Developed and deployed a web application simplifying advanced database operations for technical and non-technical teams
- Refactored manufacturing services integrating PLC systems and messaging brokers, and automated documentation updates via web scraping and syntax parsers.

EDUCATION

B.S. Computer Science & Applied Math (Double) — UW Eau Claire (3.98/4.0) 09/22 - 12/24

B.S. Computer Science — University of Minnesota - Twin Cities 09/18 - 12/19

TECHNICAL SKILLS

Langs & Libraries	Python, TypeScript, PyTorch, OpenCV, Open3D, AWS CDK
Hardware	Raspberry Pi Pico 2W, Orange Pi 5B, Machine Vision Cameras (e.g. See3CAM.24CUG)
Other	AWS (e.g. SageMaker), Git/Github, Linux, CAD

HONORS AND AWARDS

Graduated with Honors in Computer Science — *UW Eau Claire* 12/24

Fred F. and Lorain E. Winrich Scholarship — *UW Eau Claire* 09/23 - 12/24

EXTRACURRICULAR & PROJECTS

President of the Students' Association for Computing Machinery — Designed and led interactive and educational events for my schools chapter of SACM. Event topics included Git/Github, Python, and Neural Networks. Led meetings and delegated assignments to board members.

Instructor for Python & Machine Learning Workshop — Designed and ran a workshop for over 20 UWEC students, on Python, statistics, data analytics, and machine learning.

Voice-Assistant Task Manager 🗣️ — A full-stack, real-time voice-assistant task manager that allows users to create, edit, and manage tasks entirely through natural language without traditional UI controls. The system streams audio to a Python agent that interprets commands with speech-to-text and LLMs, executes database operations, and synchronizes updates to a Next.js frontend in real time.